

Learning Intention: I can explain how people use scientific explanations to solve problems and meet everyday needs.

Success Criteria — I can:

- Identify a problem that a scientific explanation helped to solve.
- Explain how observing patterns in nature can help people make decisions.
- Describe a scientific solution to an everyday need.

Reading Strategy

Before reading, look at the heading and any bold words. As you read, look for the **problem** in each example, and the **scientific explanation** that helped solve it.

Read — Solving Problems with Science

People use scientific explanations to solve real problems and meet everyday needs — this has been happening for a very long time, in many different ways.

For thousands of years, First Nations Australians have carefully observed the natural world — noticing when certain plants flower, when animals migrate, and when other seasonal changes happen. This knowledge, built up and passed down over generations, helps answer important questions, like the best time to gather certain foods or when particular animals will be found in an area. It is a deep and detailed scientific understanding of Country, developed through careful long-term observation.

Scientists today use a similar approach. By understanding that invisible ultraviolet (UV) rays from the sun can damage skin, scientists were able to develop **sunscreen** — a solution that helps protect people outdoors. By understanding that tiny living things called germs can make people sick, scientists developed **water filters** that remove germs, making water safer to drink.

In each of these examples, careful observation led to a scientific explanation, and that explanation was used to solve a real problem or meet a genuine need.

Vocabulary Helper

Word	Meaning
Observation	Carefully watching and noticing something
Traditional knowledge	Understanding built up and passed down over many generations
Solution	A way of solving a problem or meeting a need

Parent Tip

Talk together about a problem in your own life that science has helped solve — from band-aids for cuts to fridges for keeping food fresh. Ask "What did people need to understand first, before they could invent that?"

Did You Know?

First Nations Australians hold some of the oldest continuous knowledge systems in the world, developed and refined over tens of thousands of years of careful observation of plants, animals, weather and land.

Find It, Think About It & Apply

Q1 — Find It (Literal)

a) What have First Nations Australians carefully observed for thousands of years?

b) What did understanding UV rays help scientists develop? _____

c) What did understanding germs help scientists develop? _____

Q2 — Think About It (Inferred)

a) Why is knowing when plants flower or animals migrate useful for finding food?

b) Why did scientists need to understand germs before they could invent a water filter?

Q3 — Match the Problem to the Solution

Problem	Scientific solution
UV rays can damage skin	
Germs in water can make people sick	
Knowing the best time to find food in nature	

Q4 — Observation or Not?

Statement	Observation / Not an Observation
"I noticed this plant flowers every spring."	
"I just feel like something will happen today."	
"I've counted this bird visiting the garden every morning."	

Watch Out!

Traditional knowledge built over generations of careful observation is genuine scientific understanding — it isn't just an old story or guess. Long-term, careful observation is the same process scientists use today.

Q5 — True or False?

Statement	True / False
Traditional First Nations knowledge is based on careful, long-term observation of nature.	True / False
Scientific explanations are only useful for inventing new machines.	True / False

Apply & Reasoning — Use what you've read to explain your thinking.

- 1 Reasoning:** Why did scientists need to understand UV rays before they could invent sunscreen?

- 2 Reasoning:** Explain how long-term observation of nature can be considered a genuine scientific process.

- 3 Open-ended:** Think of a problem in your own life. Describe what a person would need to observe or understand scientifically before they could solve it, and describe your idea for a solution.

Quick Check — Exit Ticket

In one sentence, explain how careful observation can lead to a scientific solution.

Answer & Teacher Support

Answer Key — Pages 1 & 2

Q1 (p2): a) When certain plants flower, when animals migrate, and other seasonal changes. b) Sunscreen. c) Water filters.

Q2 (p2): a) Noticing seasonal patterns tells people the best time certain foods will be available. b) Understanding that germs cause sickness showed scientists what needed to be removed from water to make it safe.

Q3 (p2): UV rays damage skin — Sunscreen. Germs in water — Water filters. Best time to find food — Traditional ecological knowledge/seasonal observation.

Q4 (p2): "Noticed plant flowers every spring" — Observation. "Just feel like something will happen" — Not an observation. "Counted bird visiting every morning" — Observation.

Q5 (p2): "Traditional knowledge based on careful observation" — True. "Science only useful for machines" — False.

Answer Key — Page 3

Q1: Without understanding that UV rays cause skin damage, scientists wouldn't have known what problem needed solving, or what an effective solution needed to block.

Q2: It involves careful, repeated observation over time, leading to reliable explanations and predictions — the same process modern scientists use.

Q3 (acceptable-answer guide): Accept any realistic problem with a plausible scientific understanding required and a genuine, sensible proposed solution.

Exit ticket: Accept any answer showing that noticing patterns through observation can lead to an explanation, which can then be used to solve a problem.

Teacher Checklist

- ☐ Retrieves literal facts from the nonfiction text.
- ☐ Makes reasonable inferences beyond the text.
- ☐ Matches a scientific problem to its solution.
- ☐ Recognises traditional ecological knowledge as genuine science.
- ☐ Proposes a realistic scientific solution to a self-chosen problem.

Teacher Tip

This resource combines nonfiction reading comprehension with AC9S3H02 (using scientific explanations to solve problems) — a genuine curriculum gap this fills, including the curriculum's suggested recognition of First Nations ecological knowledge. Handle this content respectfully and factually, avoiding overgeneralisation about specific cultural groups.